Numerical Integration

Chapter 5 of Numerical Analysis

Yiping Cheng
ypcheng@bjtu.edu.cn
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School of Electronic and Information Engineering
Beijing Jiaotong University

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Elements of Numerical Integration

Numerical integration and quadrature

In this chapter, the terms "**numerical integration**" and "**quadrature**" are used interchangeably, both referring to the numerical approximation of definite integrals.

In numerical integration, we use a sum $\sum_{i=0}^{n} w_i f(x_i)$, called **quadrature formula**, or a quadrature rule, or simply a rule, to approximate $\int_a^b f(x)dx$. As in the last chapter, the points x_i are called nodes. And the w_i are called weights.

Now the questions arise:

- How to select a set of distinct nodes $x_0, x_1, \cdots, x_n$?
- How to choose the weights $w_0, w_1, \cdots, w_n$?

Algebraic accuracy of quadrature formulas

Definition (algebraic accuracy)

The algebraic accuracy of a quadrature formula $\sum_{i=0}^{n} w_i f(x_i)$ for the integral $\int_a^b f(x)dx$ is the smallest integer p such that the formula is not exact when $f(x) = x^p$.

Some remarks:

- Left-hand rectangle rule $(b-a)f(a)$ has algebraic accuracy 1.
- Midpoint rule $(b-a)f(\frac{a+b}{2})$ has algebraic accuracy 2.
- Trapezoid rule $(b-a)\frac{f(a)+f(b)}{2}$ has algebraic accuracy 2.
- Algebraic accuracy is an indirect, and only one of the many quality measures of quadrature formulas.

Example

Find the algebraic accuracy of $\int_{-1}^{1} f(x)dx \approx \frac{1}{2}[f(-1) + 2f(0) + f(1)]$.

Solution:

$$\int_{-1}^{1} x^0 dx = 2, \text{ and } \frac{1}{2}[1 + 2 * 1 + 1] = 2$$

$$\int_{-1}^{1} x^1 dx = 0, \text{ and } \frac{1}{2}[-1 + 2 * 0 + 1] = 0$$

$$\int_{-1}^{1} x^2 dx = \frac{2}{3}, \text{ but } \frac{1}{2}[1 + 2 * 0 + 1] = 1.$$

Hence the algebraic accuracy of this formula is 2.

Equivalent definition of algebraic accuracy

The previous definition has an equivalent version that will be more convenient in later derivations.

Proposition

The algebraic accuracy of a quadrature formula $\sum_{i=0}^{n} w_i f(x_i)$ for the integral $\int_a^b f(x)dx$ is the smallest integer p such that the formula is not exact when $f(x) = (x-a)^p$.

In the sequel, let us denote

$$\tilde{x}_i = x_i - a, \text{ for } i = 0, 1, \ldots, n.$$

Maximum algebraic accuracy with $n+1$ free nodes

To satisfy the algebraic accuracy p requirement the following equations must be fulfilled:

$$
\begin{aligned}
w_0 + w_1 + \cdots + w_n &= b - a \\
w_0\tilde{x}_0 + w_1\tilde{x}_1 + \cdots + w_n\tilde{x}_n &= \frac{(b-a)^2}{2} \\
&\vdots \\
w_0\tilde{x}_0^{p-1} + w_1\tilde{x}_1^{p-1} + \cdots + w_n\tilde{x}_n^{p-1} &= \frac{(b-a)^p}{p}.
\end{aligned}
$$

There are altogether p equations. If both the w_i and the x_i are treated as unknowns, then we have $2n+2$ unknowns. Therefore, we can expect an algebraic accuracy of $2n+1$ with $n+1$ free nodes. If a quadrature formula with $n+1$ nodes has algebraic accuracy $2n+1$, then it is called a **Gaussian quadrature**.

The discussion of Gaussian quadratures is omitted here due to space and time limitation.

Algebraic accuracy with $n+1$ prescribed nodes

If the $n+1$ nodes $x_0, x_1, \cdots, x_n$ are chosen beforehand, then we only have $n+1$ unknowns $w_0, w_1, \cdots, w_n$, and we can have $n+1$ equations:

$$
\begin{aligned}
w_0 + w_1 + \cdots + w_n &= b - a \\
w_0\tilde{x}_0 + w_1\tilde{x}_1 + \cdots + w_n\tilde{x}_n &= \frac{(b-a)^2}{2} \\
&\vdots \\
w_0\tilde{x}_0^n + w_1\tilde{x}_1^n + \cdots + w_n\tilde{x}_n^n &= \frac{(b-a)^{n+1}}{n+1}.
\end{aligned}
$$

It is assumed that $a \le x_0 < x_1 < \cdots < x_n \le b$. Then the above system of equations with $w_0, w_1, \cdots, w_n$ as unknowns has a unique solution, which ensures the unique existence of a quadrature formula with **algebraic accuracy at least** $n+1$.

Interpolatory quadrature formulas

Suppose we are given nodes $a \le x_0 < x_1 < \cdots < x_n \le b$, and we want to have a quadrature formula with algebraic accuracy at least $n+1$. It is tantamount to solving the system of equations given in the previous slide. The system can be rewritten in matrix form:

$$\begin{bmatrix} 1 & 1 & \cdots & 1 \\ \tilde{x}_0 & \tilde{x}_1 & \cdots & \tilde{x}_n \\ \vdots & \vdots & \ddots & \vdots \\ \tilde{x}_0^n & \tilde{x}_1^n & \cdots & \tilde{x}_n^n \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_n \end{bmatrix} = \begin{bmatrix} b-a \\ \frac{(b-a)^2}{2} \\ \vdots \\ \frac{(b-a)^{n+1}}{n+1} \end{bmatrix}.$$

Thus,

$$\begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_n \end{bmatrix} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ \tilde{x}_0 & \tilde{x}_1 & \cdots & \tilde{x}_n \\ \vdots & \vdots & \ddots & \vdots \\ \tilde{x}_0^n & \tilde{x}_1^n & \cdots & \tilde{x}_n^n \end{bmatrix}^{-1} \begin{bmatrix} b-a \\ \frac{(b-a)^2}{2} \\ \vdots \\ \frac{(b-a)^{n+1}}{n+1} \end{bmatrix}.$$

$$w_i = \sum_{k=0}^{n} \frac{(-1)^{n-k}\sigma_{n-k}(\{\tilde{x}_0, \tilde{x}_1, \cdots, \tilde{x}_n\} \backslash \{\tilde{x}_i\})}{\prod_{0 \le j \le n, j \ne i}(\tilde{x}_i - \tilde{x}_j)} \frac{(b-a)^{k+1}}{k+1}.$$

The previous equation for the weights have an equivalent expression that can be derived as follows.

$$\begin{bmatrix} 1 & 1 & \cdots & 1 \\ x_0 & x_1 & \cdots & x_n \\ \vdots & \vdots & \ddots & \vdots \\ x_0^n & x_1^n & \cdots & x_n^n \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_n \end{bmatrix} = \begin{bmatrix} b-a \\ \frac{b^2-a^2}{2} \\ \vdots \\ \frac{b^{n+1}-a^{n+1}}{n+1} \end{bmatrix}.$$

Thus,

$$\begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_n \end{bmatrix} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ x_0 & x_1 & \cdots & x_n \\ \vdots & \vdots & \ddots & \vdots \\ x_0^n & x_1^n & \cdots & x_n^n \end{bmatrix}^{-1} \begin{bmatrix} b-a \\ \frac{b^2-a^2}{2} \\ \vdots \\ \frac{b^{n+1}-a^{n+1}}{n+1} \end{bmatrix} = \begin{bmatrix} c_{0,0} & c_{0,1} & \cdots & c_{0,n} \\ c_{1,0} & c_{1,1} & \cdots & c_{1,n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n,0} & c_{n,1} & \cdots & c_{n,n} \end{bmatrix} \begin{bmatrix} b-a \\ \frac{b^2-a^2}{2} \\ \vdots \\ \frac{b^{n+1}-a^{n+1}}{n+1} \end{bmatrix}.$$

$$w_i = \sum_{k=0}^{n} c_{i,k} \frac{b^{k+1}-a^{k+1}}{k+1} = \sum_{k=0}^{n} \frac{(-1)^{n-k}\sigma_{n-k}(\{x_0, x_1, \cdots, x_n\}\backslash\{x_i\})}{\prod_{0\le j\le n, j\ne i}(x_i - x_j)} \frac{b^{k+1}-a^{k+1}}{k+1}.$$

Why "interpolatory quadrature formulas"?

The above derived quadrature formulas are traditionally called **interpolatory quadrature formulas**. Why the name? By "interpolatory" we mean that we approximate $\int_a^b f(x)dx$ by $\int_a^b p(x)dx$ where $p(x)$ is the Lagrange interpolating polynomial passing through $(x_0, f(x_0)), (x_1, f(x_1)), \cdots, (x_n, f(x_n))$. We have known in Chapter 4 that $p(x) = \sum_{i=0}^n f(x_i)\ell_i(x)$ where

$$\ell_i(x) = \prod_{0\le j\le n, j\ne i} \frac{x - x_j}{x_i - x_j}.$$

We then have $\int_a^b p(x)dx = \sum_{i=0}^n f(x_i) \int_a^b \ell_i(x)dx$ which means $w_i = \int_a^b \ell_i(x)dx$ in this quadrature formula. Further,

$$w_i = \int_a^b \ell_i(x)dx = \int_a^b \sum_{k=0}^n c_{i,k}x^k dx = \sum_{k=0}^n c_{i,k} \int_a^b x^k dx = \sum_{k=0}^n c_{i,k} \frac{b^{k+1} - a^{k+1}}{k+1},$$

which agrees with the algebraic accuracy at least $n + 1$ formula given in the previous slide.

Stability of quadrature formulas

Definition (stable quadrature formula)

A quadrature formula $\sum_{i=0}^{n} w_i f(x_i)$ is said to be stable, if all $w_i > 0$.

Consider the errors in computing the $f(x_i)$ whose absolute values are bounded by δ, and the obtained quadrature is then $\sum_{i=0}^{n} w_i f_i$ with $|f_i - f(x_i)| \leq \delta$. Then the total absolute error

$$|\sum_{i=0}^{n} w_i f_i - \sum_{i=0}^{n} w_i f(x_i)| = |\sum_{i=0}^{n} w_i [f_i - f(x_i)]| \leq \sum_{i=0}^{n} |w_i| \delta.$$

Suppose the formula has algebraic accuracy at least 1, and is stable, i.e. all weights $w_i > 0$. Then $\sum_{i=0}^{n} w_i = b - a$ and $|w_i| = w_i$ and the above total absolute error is bounded by

$$\sum_{i=0}^{n} |w_i| \delta = \sum_{i=0}^{n} w_i \delta = (b - a)\delta.$$

If the formula is not stable, then such a bound is not easily found.

Newton-Cotes Formulas

Definitions of Newton-Cotes formulas

Definition

The **closed Newton-Cotes formula of grid-order** n is the interpolatory quadrature formula with closed equally-spaced $n+1$ nodes, i.e. $x_i = a + ih$, $h = \frac{b-a}{n}$, $i = 0, 1, \ldots, n$.

Definition

The **open Newton-Cotes formula of grid-order** n is the interpolatory quadrature formula with open equally-spaced $n-1$ nodes, i.e. $x_i = a + ih$, $h = \frac{b-a}{n}$, $i = 1, 2, \ldots, n-1$.

By **closed** we mean that the nodes include the two endpoints, and **open** the contrary. Here we use the word "grid-order" to distinguish it from "order" which is regarded as a synonym of "algebraic accuracy".

Closed Newton-Cotes formulas in general form

Now since $h = \frac{b-a}{n}$ and $x_i = a + ih$, for $i = 0, 1, \ldots, n$, we have

$$
\begin{aligned}
w_i &= \sum_{k=0}^{n} \frac{(-1)^{n-k}\sigma_{n-k}(\{\tilde{x}_0, \tilde{x}_1, \cdots, \tilde{x}_n\}\backslash\{\tilde{x}_i\})}{\prod_{0\le j\le n, j\ne i}(\tilde{x}_i - \tilde{x}_j)} \frac{(b-a)^{k+1}}{k+1} \\
&= \sum_{k=0}^{n} \frac{(-1)^{n-k}h^{n-k}\sigma_{n-k}(\{0, 1, \ldots, n\}\backslash\{i\})}{h^n \prod_{0\le j\le n, j\ne i}(i-j)} \frac{(nh)^{k+1}}{k+1} \\
&= (b-a) \underbrace{\frac{\sum_{k=0}^{n} \frac{(-1)^{n-k}n^k\sigma_{n-k}(\{0,1,\ldots,n\}\backslash\{i\})}{k+1}}{\prod_{0\le j\le n, j\ne i}(i-j)}}_{C_i^{(n)}}.
\end{aligned}
$$

Table of closed Newton-Cotes coefficients

n	$C_0^{(n)}$	$C_1^{(n)}$	$C_2^{(n)}$	$C_3^{(n)}$	$C_4^{(n)}$	$C_5^{(n)}$	$C_6^{(n)}$	$C_7^{(n)}$	$C_8^{(n)}$
1	$\frac{1}{2}$	$\frac{1}{2}$							
2	$\frac{1}{6}$	$\frac{4}{6}$	$\frac{1}{6}$						
3	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$					
4	$\frac{7}{90}$	$\frac{32}{90}$	$\frac{12}{90}$	$\frac{32}{90}$	$\frac{7}{90}$				
5	$\frac{19}{288}$	$\frac{75}{288}$	$\frac{50}{288}$	$\frac{50}{288}$	$\frac{75}{288}$	$\frac{19}{288}$			
6	$\frac{41}{840}$	$\frac{216}{840}$	$\frac{27}{840}$	$\frac{272}{840}$	$\frac{27}{840}$	$\frac{216}{840}$	$\frac{41}{840}$		
7	$\frac{751}{17280}$	$\frac{3577}{17280}$	$\frac{1323}{17280}$	$\frac{2989}{17280}$	$\frac{2989}{17280}$	$\frac{1323}{17280}$	$\frac{3577}{17280}$	$\frac{751}{17280}$	
8	$\frac{989}{28350}$	$\frac{5888}{28350}$	$-\frac{928}{28350}$	$\frac{10496}{28350}$	$-\frac{4540}{28350}$	$\frac{10496}{28350}$	$-\frac{928}{28350}$	$\frac{5888}{28350}$	$\frac{989}{28350}$

Open Newton-Cotes formulas in general form

Now since $h = \frac{b-a}{n}$ and $x_i = a + ih,$ for $i = 1, 2, \ldots, n-1$, we have

$$
\begin{aligned}
w_i &= \sum_{k=1}^{n-1} \frac{(-1)^{n-2-k}\sigma_{n-2-k}(\{\tilde{x}_1, \tilde{x}_2, \cdots, \tilde{x}_{n-1}\}\backslash\{\tilde{x}_i\})}{\prod_{1\le j\le n-1, j\ne i}(\tilde{x}_i - \tilde{x}_j)} \frac{(b-a)^{k+1}}{k+1} \\
&= \sum_{k=1}^{n-1} \frac{(-1)^{n-2-k} h^{n-2-k}\sigma_{n-2-k}(\{1, 2, \ldots, n-1\}\backslash\{i\})}{h^{n-2}\prod_{1\le j\le n-1, j\ne i}(i-j)} \frac{(nh)^{k+1}}{k+1} \\
&= (b-a) \underbrace{\frac{\sum_{k=1}^{n-1} \frac{(-1)^{n-2-k} n^k \sigma_{n-2-k}(\{1,2,\ldots,n-1\}\backslash\{i\})}{k+1}}{\prod_{1\le j\le n-1, j\ne i}(i-j)}}_{O_i^{(n)}}.
\end{aligned}
$$

Table of open Newton-Cotes coefficients

n	$O_1^{(n)}$	$O_2^{(n)}$	$O_3^{(n)}$	$O_4^{(n)}$	$O_5^{(n)}$
2	1				
3	$\frac{1}{2}$	$\frac{1}{2}$			
4	$\frac{2}{3}$	$-\frac{1}{3}$	$\frac{2}{3}$		
5	$\frac{11}{24}$	$\frac{1}{24}$	$\frac{1}{24}$	$\frac{11}{24}$	
6	$\frac{11}{20}$	$-\frac{14}{20}$	$\frac{26}{20}$	$-\frac{14}{20}$	$\frac{11}{20}$

Summary of closed Newton-Cotes rules

Some closed Newton-Cotes rules with error terms are as follows. Here, $a = x_0$, $b = x_n$, $h = (b-a)/n$, $x_i = a + ih$, $f_i = f(x_i)$, for $i = 0, 1, \ldots, n$, and $a = x_0 < \xi < x_n = b$ in the error terms.

Trapezoid rule:

$$\int_{x_0}^{x_1} f(x)dx = \frac{h}{2}(f_0 + f_1) - \frac{1}{12}h^3 f''(\xi).$$

Simpson's rule:

$$\int_{x_0}^{x_2} f(x)dx = \frac{h}{3}(f_0 + 4f_1 + f_2) - \frac{1}{90}h^5 f^{(4)}(\xi).$$

Simpson's 3/8 rule:

$$\int_{x_0}^{x_3} f(x)dx = \frac{3h}{8}(f_0 + 3f_1 + 3f_2 + f_3) - \frac{3}{80}h^5 f^{(4)}(\xi).$$

Boole's rule:

$$\int_{x_0}^{x_4} f(x)dx = \frac{2h}{45}(7f_0 + 32f_1 + 12f_2 + 32f_3 + 7f_4) - \frac{8}{945}h^7 f^{(6)}(\xi).$$

Six-point Newton-Cotes closed rule:

$$\int_{x_0}^{x_5} f(x)dx = \frac{5h}{288}(19f_0 + 75f_1 + 50f_2 + 50f_3 + 75f_4 + 19f_5) - \frac{275}{12096}h^7 f^{(6)}(\xi).$$

Summary of open Newton-Cotes rules

Some open Newton-Cotes rules with error terms are as follows. Here, $a = x_0$, $b = x_n$, $h = (b-a)/n$, $x_i = a + ih$, $f_i = f(x_i)$, for $i = 1, \ldots, n-1$, and $a = x_0 < \xi < x_n = b$ in the error terms.

Midpoint rule:

$$\int_{x_0}^{x_2} f(x)dx = 2hf_1 + \frac{1}{24}h^3 f''(\xi).$$

Two-point Newton-Cotes open rule:

$$\int_{x_0}^{x_3} f(x)dx = \frac{3h}{2}(f_1 + f_2) + \frac{1}{4}h^3 f''(\xi).$$

Three-point Newton-Cotes open rule:

$$\int_{x_0}^{x_4} f(x)dx = \frac{4h}{3}(2f_1 - f_2 + 2f_3) + \frac{28}{90}h^5 f^{(4)}(\xi).$$

Four-point Newton-Cotes open rule:

$$\int_{x_0}^{x_5} f(x)dx = \frac{5h}{24}(11f_1 + f_2 + f_3 + 11f_4) + \frac{95}{144}h^5 f^{(4)}(\xi).$$

Five-point Newton-Cotes open rule:

$$\int_{x_0}^{x_6} f(x)dx = \frac{6h}{20}(11f_1 - 14f_2 + 26f_3 - 14f_4 + 11f_5) - \frac{41}{140}h^7 f^{(6)}(\xi).$$

Composite Quadrature

The need for composite quadrature rules

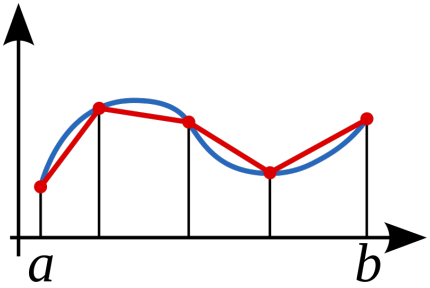
So far, we have fit one interpolant to the entire interval $[a, b]$ and obtain a quadrature rule. One way to derive quadrature rules of higher and higher order is to fit polynomial interpolants of higher and higher degree to the integrand $f(x)$ and integrate these interpolants exactly. However, as we know, high-degree polynomial interpolation based on equally spaced abscissae generically results in oscillatory interpolants that do not reflect the behaviour of the underlying function. This would degrade the accuracy of the quadrature rules based on them.

This section describes an alternative approach: Break the interval $[a, b]$ into subintervals, and apply the low-order Newton-Cotes rules on each subinterval or combination of consecutive subintervals. This approach is called **composite quadrature**.

Composite Trapezoid rule

Divide the interval $[a, b]$ into n equal-length subintervals $[x_{i-1}, x_i]$, with $h = \frac{b-a}{n}$, $x_i = a + ih$ for $i = 0, 1, \ldots, n$. Apply the Trapezoid rule to each subinterval. Then

$$\begin{aligned}\int_a^b f(x)dx &= \sum_{i=1}^{n} \int_{x_{i-1}}^{x_i} f(x)dx \\ &\approx \sum_{i=1}^{n} \frac{h}{2}[f(x_{i-1}) + f(x_i)] \\ &= h[\frac{f(a) + f(b)}{2} + \sum_{i=1}^{n-1} f(x_i)].\end{aligned}$$



Error term of composite Trapezoid rule

Theorem

Suppose f has continuous 2nd derivative, there is $\xi \in (a, b)$ such that

$$\int_a^b f(x)dx = h[\frac{f(a)+f(b)}{2} + \sum_{i=1}^{n-1} f(x_i)] - \frac{(b-a)h^2}{12} f''(\xi).$$

Proof.

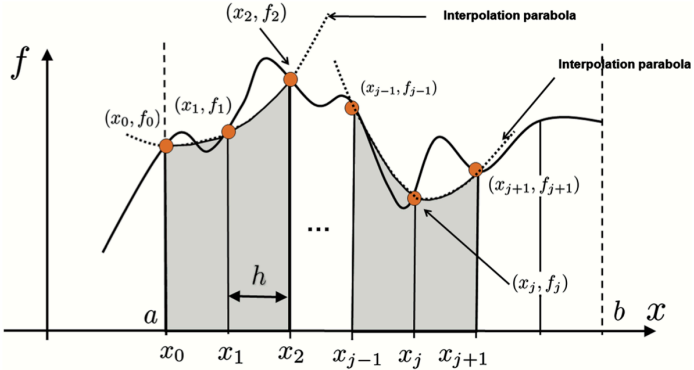
For $i = 1, \ldots, n$, $\int_{x_{i-1}}^{x_i} f(x)dx = \frac{h}{2}[f(x_{i-1} + f(x_i)] - \frac{1}{12}h^3 f''(\xi_i)$ where $x_{i-1} < \xi_i < x_i$.
Adding them together, we obtain

$$\int_a^b f(x)dx = h[\frac{f(a)+f(b)}{2} + \sum_{i=1}^{n-1} f(x_i)] - \sum_{i=1}^{n} \frac{1}{12}h^3 f''(\xi_i).$$

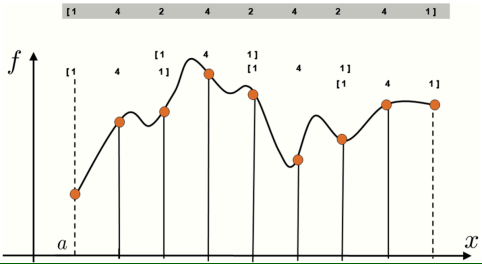
By the intermediate value theorem, there is $\xi \in (a, b)$ such that $f''(\xi) = \frac{1}{n}\sum_{i=1}^{n} f''(\xi_i)$. Hence the result. □

Composite Simpson's rule

Divide the interval $[a, b]$ into an **even number** n of equal-length subintervals $[x_{i-1}, x_i]$, with $h = \frac{b-a}{n}$, $x_i = a + ih$ for $i = 0, 1, \ldots, n$. Apply Simpson's rule to the consecutive intervals $[x_{2j-2}, x_{2j-1}]$ and $[x_{2j-1}, x_{2j}]$, for $j = 1, \ldots, \frac{n}{2}$. Then



$$
\begin{aligned}
\int_a^b f(x)dx &= \sum_{j=1}^{\frac{n}{2}} \int_{x_{2j-2}}^{x_{2j}} f(x)dx \\
&\approx \sum_{j=1}^{\frac{n}{2}} \frac{h}{3}[f(x_{2j-2}) + 4f(x_{2j-1}) + f(x_{2j})] \\
&= \frac{h}{3}\left(f(a) + f(b) + \sum_{j=1}^{\frac{n}{2}-1} [4f(x_{2j-1}) + 2f(x_{2j})] \right).
\end{aligned}
$$



Error term of composite Simpson's rule

Theorem

Suppose f has continuous 4th derivative, there is $\xi \in (a, b)$ such that

$$\int_a^b f(x)dx = \frac{h}{3}\left(f(a) + f(b) + \sum_{j=1}^{\frac{n}{2}-1}[4f(x_{2j-1}) + 2f(x_{2j})]\right) - \frac{(b-a)h^4}{180}f^{(4)}(\xi).$$

Proof.

For $j = 1, \ldots, \frac{n}{2}$, $\int_{x_{2j-2}}^{x_{2j}} f(x)dx = \frac{h}{3}[f(x_{2j-2}) + 4f(x_{2j-1}) + f(x_{2j})] - \frac{1}{90}h^5 f^{(4)}(\xi_i)$ where $x_{2j-2} < \xi_i < x_{2j}$.

Adding them together, we obtain

$$\int_a^b f(x)dx = \frac{h}{3}\left(f(a) + f(b) + \sum_{j=1}^{\frac{n}{2}-1}[4f(x_{2j-1}) + 2f(x_{2j})]\right) - \sum_{j=1}^{\frac{n}{2}} \frac{1}{90}h^5 f^{(4)}(\xi_j).$$

By the intermediate value theorem, there is $\xi \in (a, b)$ such that $f^{(4)}(\xi) = \frac{2}{n}\sum_{j=1}^{\frac{n}{2}} f^{(4)}(\xi_j)$. Hence the result. □

Adaptive Quadrature

Introducing adaptive quadrature

Adaptive quadrature involves careful selection of the points where $f(x)$ is sampled. We want to evaluate the function at as few points as possible while approximating the integral to within some specified accuracy. A fundamental additive property of a definite integral is the basis for adaptive quadrature. If c is any point between a and b, then

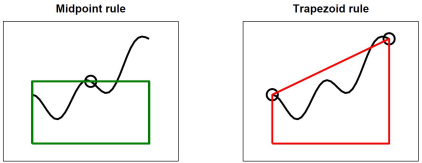
$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx.$$

The idea is that if we can approximate each of the two integrals on the right to within a specified tolerance, then the sum gives us the desired result. If not, we can recursively apply the additive property to each of the intervals $[a, c]$ and $[c, b]$. The resulting algorithm will adapt to the integrand automatically, partitioning the interval into subintervals with fine spacing where the integrand is varying rapidly and coarse spacing where the integrand is varying slowly.

Midpoint and trapezoid rules

The derivation of our adaptive quadrature rule begins with two of the basic quadrature rules shown in the following figure: the midpoint rule and the trapezoid rule. Let $h = b - a$ be the length of the interval. The midpoint rule, M, approximates the integral by the area of a rectangle whose base has length h and whose height is the value of the integrand at the midpoint:

$$M = hf(\frac{a+b}{2}).$$



The trapezoid rule, T, approximates the integral by the area of a trapezoid with base h and sides equal to the values of the integrand at the two endpoints:

$$T = h\frac{f(a)+f(b)}{2}.$$

The accuracy of a quadrature rule can be predicted in part by examining its behavior on polynomials. The order of a quadrature rule is the degree of the lowest degree polynomial that the rule does not integrate exactly. If a quadrature rule of order p is used to integrate a smooth function over a small interval of length h, then a Taylor series analysis shows that the error is proportional to h^p. The midpoint rule and the trapezoid rule are both exact for constant and linear functions of x, but neither of them is exact for a quadratic in x, so they both have order two. (The order of a rectangle rule with height $f(a)$ or $f(b)$ instead of the midpoint is only one.)

The accuracy of the two rules can be compared by examining their behavior on the simple integral

$$\int_0^1 x^2 dx = \frac{1}{3}.$$

The midpoint rule gives $M = 1(\frac{1}{2})^2 = \frac{1}{4}$. The trapezoid rule gives $T = 1(\frac{0^2+1^2}{2}) = \frac{1}{2}$. So the error in M is $\frac{1}{12}$, while the error in T is $\frac{1}{6}$. The errors have opposite signs and, perhaps surprisingly, the midpoint rule is twice as accurate as the trapezoid rule.

Simpson's rule

This turns out to be true more generally. For integrating smooth functions over short intervals, M is roughly twice as accurate as T and the errors have opposite signs. Knowing these error estimates allows us to combine the two and get a rule that is usually more accurate than either one separately. If the error in T were exactly -2 times the error in M, then solving

$$S - T = -2(S - M)$$

for S would give us the exact value of the integral. In any case, the solution

$$S = \frac{2}{3}M + \frac{1}{3}T$$

is usually a more accurate approximation than either M or T alone. This rule is known as Simpson's rule. It can also be derived by integrating the quadratic function that interpolates the integrand at the two endpoints, a and b, and the midpoint, $c = \frac{a+b}{2}$:

$$S = \frac{h}{6}(f(a) + 4f(c) + f(b)).$$

It turns out that S also integrates cubics exactly, but not quartics, so its order is four.

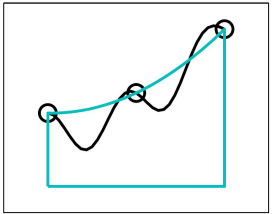
Composite Simpson's rule

We can carry this process one step further using the two halves of the interval, $[a, c]$ and $[c, b]$. Let d and e be the midpoints of these two subintervals: $d = \frac{a+c}{2}$ and $e = \frac{c+b}{2}$. Apply Simpson's rule to each subinterval to obtain a quadrature rule over $[a, b]$:

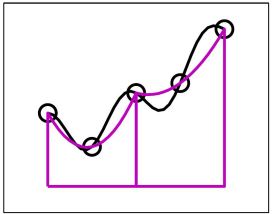
$$S_2 = \frac{h}{12}(f(a) + 4f(d) + 2f(c) + 4f(e) + f(b)).$$

This is an example of a composite quadrature rule.

Simpson's rule



Composite Simpson's rule



Boole's rule

S and S_2 approximate the same integral, so their difference can be used as an estimate of the error:

$$E = S_2 - S.$$

Moreover, the two can be combined to get an even more accurate approximation, Q. Both rules are of order four, but the S_2 step size is half the S step size, so S_2 is roughly 2^4 times as accurate. Thus, Q is obtained by solving

$$Q - S = 16(Q - S_2).$$

The result is

$$Q = \frac{16S_2 - S}{15}.$$

You are asked to express Q as a weighted combination of the five function values $f(a)$ through $f(e)$ and to establish that its order is six. The rule is known as Boole's rule, the sixth-order Newton-Cotes rule. We will simply call it the extrapolated Simpson's rule because it uses Simpson's rule for two different values of h and then extrapolates toward $h = 0$.